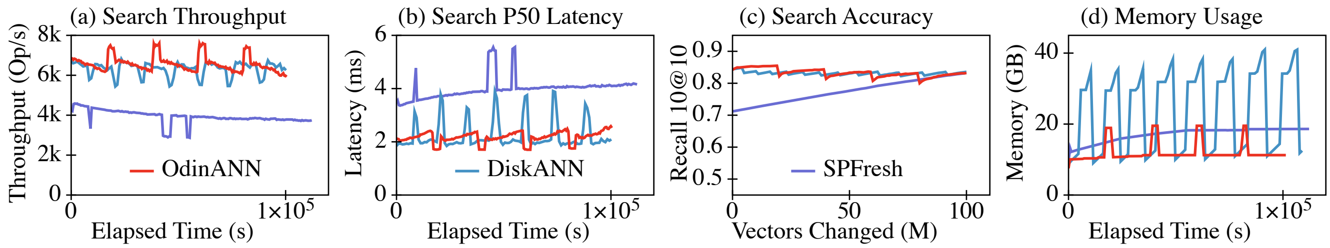

Figure 12: Workload with deletes in SIFT100M dataset, replace the original 100M vectors with another 100M vectors. Note that OdinANN merges at 20% updated vectors, while DiskANN merges at 6% for less memory usage. With similar elapsed time in this figure, OdinANN and SPFresh finish replacing the whole 100M vectors, while DiskANN replaces 50M vectors.

5 Related Work

Billion-Scale ANNS indexes. Real-world workloads do vector search on billions of vectors [7, 22]. Traditional ANNS indexes are stored in memory [4, 7, 18]. To support billion-scale datasets in a single node, they suffer from high prices and limited scalability. To this end, two new methods, *hybrid storage* and *vector compression*, are proposed.

Hybrid storage. Some works rely on hybrid storage media for more capacity, such as SSD and DRAM. DiskANN [24] and Starling [27] are on-disk graph-based indexes. They store the full graph on disk, and compressed vectors in memory for graph navigation. SPANN [5] is an on-disk cluster-based index. It stores the clusters on disk, and a small graph in memory to index the nearest clusters. HM-ANN [21] maintains a hierarchical graph [18] in DRAM and persistent memory.

Some other works seek performance boosts with near-data processing. CXL-ANNS [11] uses processing units near the CXL memory for distance calculation, and SmartANNS [25] uses FPGAs in SmartSSDs for graph traversal. These methods can not only reduce the CPU burden for computation, but mitigate the bandwidth bottleneck by avoiding transferring vectors between CPU and slow-speed storage media.

OdinANN uses commodity SSDs for capacity expansion. It does not rely on near-data processing for better compatibility.

Vector compression. Instead of increasing capacity, some works compress vectors for smaller index sizes in memory. Product quantization (PQ) [8, 14] is a popular vector compression method for ANNS. It splits vectors into equally sized chunks and does clustering for each chunk. Vectors are replaced with per-chunk IDs representing their nearest centroids. This way is used by IVF [2] indexes for less memory footprint. DiskANN [24] also stores PQ-compressed vectors in memory to reduce neighbor reads during graph navigation. LVQ [1] focuses on single-vector compression instead of cross-vector clustering. It uses scalar quantization in each vector to map floating-point vectors to bits, which improves search performance with reduced memory footprint.

OdinANN stores PQ-compressed vectors in memory for graph navigation. The designs in OdinANN are orthogonal to vector compression methods.

Updatable ANNS indexes. Graph-based DiskANN [23] uses buffered inserts, which interfere with frontend search tasks, motivating our designs for OdinANN. Cluster-based SPFresh [30] supports in-place vector updates. It is more challenging for graph-based OdinANN to support efficient updates than cluster-based SPFresh, due to the intrinsic complexity of graph neighbor maintenance. However, OdinANN benefits from better search accuracy and performance.

Reserving space for efficient updates. ALEX [6] is a learned index that also leverages reserved space (called "gaps" in its paper) for efficient updates. Unlike ALEX, which uses one gap for each new key-value pair to avoid model retraining, OdinANN uses multiple record slots for a single insert, combining record updates in the same page.

6 Conclusion

We argue that direct insert is more suitable for graph-based ANNS indexes and propose OdinANN. It efficiently processes inserts in the on-disk index instead of buffering them. Evaluations show that OdinANN keeps stable insert and search performance in billion-scale vector datasets. This work demonstrates that on-disk graph-based ANNS indexes can benefit from high search accuracy, stable search performance, as well as efficient updates simultaneously using direct insert.

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